Task 6: Critical Evaluation of Experimental Design

6.1 Why This Task Matters Here

Everything analyzed in Tasks 3–5 is **observational data** - routinely collected FBC results, not data generated under a controlled experiment. Observational data can establish association (e.g., "women have higher anemia prevalence than men," "Institution_C shows elevated prevalence") but cannot, on its own, establish that a specific intervention (e.g., an iron supplementation program, a screening protocol change, a staffing reallocation) will actually reduce anemia burden. If the client organization wants to move from diagnosing the problem to testing solutions, a designed experiment becomes the appropriate next step. This task critically evaluates whether two classical experimental design frameworks - Completely Randomized Design (CRD) and Randomized Complete Block Design (RCBD) - would be suitable for that next phase, applied specifically to our anemia context rather than in the abstract.

6.2 Completely Randomized Design (CRD)

What it is

In a CRD, experimental units (here, patients) are assigned to treatment groups (e.g., "receives iron supplementation" vs. "standard care") **entirely at random**, with no further structuring of the randomization. It is the simplest experimental design and is the classical basis of the standard randomized controlled trial (RCT).

Situations where CRD may be useful

CRD would be appropriate for a **single-institution intervention study** - for example, a trial testing whether a new iron-folic-acid supplementation reminder protocol improves hemoglobin outcomes among diagnosed anemic patients at one hospital, where the patient population is reasonably homogeneous. This mirrors real designs in the anemia-intervention literature: a randomized controlled trial comparing alternate-day versus daily oral iron dosing for iron-deficiency anemia used simple randomization (with block randomization used only for allocation-sequence balance, not for a structural blocking factor) within a single trial population of 200

adults [1]. Similarly, a trial of intravenous iron for anemia in intensive care unit patients used straightforward 1:1 randomization within a defined, relatively homogeneous patient population [2].

Expected advantages

- **Simplicity**: straightforward to implement, analyze (standard t-tests/ANOVA apply directly), and explain to non-statistician stakeholders - a real advantage when reporting back to hospital administrators or a Board.
- **Unbiased treatment allocation**: pure randomization protects against both known and unknown confounders, in principle, given a large enough sample.
- **Flexibility**: works with any number of treatment arms (e.g., could compare "no intervention" vs. "oral iron" vs. "IV iron" simultaneously).

Limitations

- CRD assumes the patient population is **relatively homogeneous** aside from the randomized treatment. Our own descriptive and inferential findings (Tasks 3–4) show this assumption is **not realistic** for this population: HB levels and anemia prevalence vary substantially and significantly by age, sex, and their interaction (Task 4, Test G: $\eta^2 = 0.066$ for the interaction alone). If these factors are not accounted for in the design, they become **noise that inflates the variance of the treatment-effect estimate**, reducing statistical power to detect a true effect - consequence of ignoring known sources of heterogeneity in design rather than controlling for them.
- With small sample sizes (plausible for early-phase trials, or if restricted to Institution_A's much smaller patient pool - 710 cases historically), pure randomization by chance can still produce imbalanced groups (e.g., more elderly patients in one arm) even though the procedure was unbiased.

Implementation challenges

- Recruiting and retaining a large enough sample within a **single institution** - the profile of feasibility risk in this literature is real, not hypothetical: a recent feasibility study of a randomized trial of IV iron after major abdominal surgery screened 205 eligible patients but recruited only 4, illustrating how

randomized designs in real clinical settings can face severe practical recruitment barriers [3].

- Ethical/practical constraints: withholding a known-effective treatment (e.g., iron supplementation for confirmed moderate/severe anemia) from a control group raises ethical questions that typically require an active-comparator design (e.g., alternate-day vs. daily dosing, as in [1]) rather than a placebo arm, once efficacy is already well-established in the literature.

Practical feasibility for this client

Feasible only for a **narrowly scoped, single-institution pilot** - for instance, testing a low-cost behavioral intervention (e.g., SMS/phone reminders to improve supplementation adherence, which showed a 44.9% vs. 13.8% adherence improvement in a comparable telephonic-intervention RCT in pregnant Indian women [4]) at the institution with the largest patient volume (Institution_B). A full multi-institution CRD would be a design mismatch given the population heterogeneity this project has already documented.

6.3 Randomized Complete Block Design (RCBD)

What it is

RCBD first groups experimental units into **blocks** of similar units (controlling for a known source of variability), then randomizes treatment assignment within each block. The logic is to remove a known nuisance factor's variability from the treatment-effect estimate rather than letting it dilute into random error, as it would under CRD.

Situations where RCBD may be useful

Given this project's own findings, **two blocking factors are directly and empirically justified**, not hypothetical:

1. **Institution as a block.** Task 4 (Test E) found a statistically significant, if modest, association between institution and anemia prevalence ($\chi^2 = 233.85$, $p < 2.2\times10^{-16}$, Cramer's V = 0.093), and the wider multi-center trial literature explicitly treats "center effects" as a real phenomenon requiring statistical handling - institutional variation is described as abounding in nearly all

clinical trials, with methods developed specifically to account for hospital/treatment-center effects in randomization and analysis [5]. A future intervention trial spanning multiple institutions should block by institution, ensuring each site contributes proportionally to both treatment and control arms, rather than risking an imbalanced split such as disproportionately more Institution_C patients landing in one arm.

2. **Age group (or age×sex stratum) as a block.** Given Task 4's confirmed age×sex interaction effect on HB, blocking by age-sex stratum before randomizing treatment would directly control for the single largest known source of baseline HB variability in this population. This mirrors real precedent: a Gambian trial of anemia interventions in pregnancy used a block design **stratified explicitly by baseline hemoglobin level and gestational stage** [6], and a trial of IV iron in ovarian cancer patients used minimization/blocking by age (<70 vs. ≥70) and surgery type specifically because these were known to affect the outcome [7].

Expected advantages

- **Increased statistical power**: by removing a known source of variance (institution, or age-sex stratum) from the error term, RCBD can detect true treatment effects with a smaller sample than CRD would require for the same power - directly relevant given the modest size of some institutions in our dataset (Institution_A, n=710).
- **Fairer, more interpretable comparisons:** guarantees every block (e.g., every institution, or every age-sex stratum) is represented in every treatment arm, rather than leaving this to chance.
- **Directly addresses a documented real-world problem**: the multi-center trial methodology literature confirms that ignoring center effects, or handling them naively, can materially affect trial power and validity, while design-based methods that explicitly incorporate center/blocking structure have been shown to meaningfully increase power in the presence of center variation [5].

Limitations

- RCBD assumes each block contains enough experimental units for every treatment arm. This is a genuine, non-trivial constraint for our specific dataset's institutional structure: with Institution_A representing only 2.6% of the historical sample, a strict RCBD blocking by all four institutions risks severely underpowered cells for that site alone, echoing a documented finding in the multi-center trial literature that adding more (smaller) centers to a trial can increase, not decrease, center-level imbalance under block-based randomization [8].
- Blocking on more than one factor simultaneously (e.g., institution and age-sex stratum together) rapidly multiplies the number of required cells, which can be impractical at typical trial sample sizes - a design trade-off, not a flaw unique to our context, but one that is more acute here given Institution_A's small historical volume.
- RCBD does not protect against confounders that were not chosen as blocking factors - e.g., if disease severity, income, or nutritional status also vary meaningfully across patients but are not blocked on, they remain a source of unexplained variance, same as under CRD.

Implementation challenges

- **Operational coordination across institutions** - a multi-site RCBD requires standardized protocols, staff training and randomization procedures replicated consistently at each site, a coordination burden not present in a single-site CRD. The clinical trials literature notes this explicitly: block randomization stratified by site can, if block sizes are chosen poorly, still yield meaningful site-level imbalance in practice, and larger block sizes or more sophisticated dynamic balancing methods are sometimes needed to manage this trade-off [8].
- **Data/administrative burden**: assigning and tracking patients by block (institution × age-sex stratum) requires more careful record-keeping than a simple CRD randomization list - a realistic operational cost for institutions already facing the resourcing constraints documented in Task 1 (e.g., limited laboratory staff availability).

Practical feasibility for this client

RCBD blocked by **institution alone** is a practically feasible and well-justified upgrade over CRD for any future multi-site anemia intervention trial this client runs - the blocking factor is not arbitrary but is drawn directly from this project's own statistical evidence (Task 4, Test E). Blocking by the full age×sex×institution cross would likely be over-ambitious given Institution_A's small patient volume; a more feasible compromise would be to block by institution only, while statistically adjusting for age and sex as covariates in the analysis stage (e.g., ANCOVA) rather than as design-stage blocks - balancing rigor against realistic recruitment constraints.

6.4 Overall Recommendation

Design	Recommended use case for this client
CRD	A small, single-institution pilot test of a low-cost, easily scaled intervention (e.g., adherence reminders), where the added complexity of blocking is not justified by the sample size or research question.
RCBD (blocked by institution)	Any future multi-institution intervention trial evaluating a treatment (e.g., supplementation protocol, screening workflow change) across this client's full network - directly motivated by this project's own Task 4 finding of a real, if modest, institution effect.

In both cases, we recommend the client treat this task's conclusions as forward-looking guidance for future primary research, not a requirement to conduct new data collection as part of this engagement. The core contribution of this section is demonstrating, using our own project's statistical findings, which design choices are justified for this specific population, rather than defaulting to generic textbook advice.

Sources cited in this section

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